

# SPACEX **FALCON 9** LANDING PREDICTION

IBM Applied Data Science Capstone

 GitHub Repository: [github.com/D8loop/spacex-landing-prediction](https://github.com/D8loop/spacex-landing-prediction)

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# EXECUTIVE SUMMARY



## The Challenge

Predicting Falcon 9 first-stage landings to determine if "Space Y" can offer competitive launch prices based on rocket reusability.



## The Methodology

End-to-end Data Science pipeline: API/Scraping, SQL analysis, Geospatial mapping (Folium), and Machine Learning classification.



## The Final Result

Achieved a prediction accuracy of 83.33% using Logistic Regression, providing a reliable baseline for financial bidding.

# DATA COLLECTION & PREPARATION

## Data Acquisition

- SpaceX REST API: Used 'requests' library to pull telemetry JSON data (filtered for Falcon 9).
- Web Scraping: Used 'BeautifulSoup' to extract historical launch records from Wikipedia HTML tables.



## Data Wrangling

- Feature Engineering: Created the binary target variable 'Class' (1 for Success, 0 for Failure).
- Data Cleaning: Handled missing values in 'PayloadMass' using mean imputation.
- Encoding: Applied One-Hot Encoding to categorical features (Orbit, Launch Site).



# EXPLORATORY DATA ANALYSIS METHODOLOGY

## Structured Exploration (SQL)

- Database Integration: Data loaded into IBM DB2 cloud environment.
- Querying Strategy: Used SQL to perform complex aggregations (SUM, AVG) and multi-table joins.
- Insights Extraction: Isolated specific success patterns per launch site and payload categories.

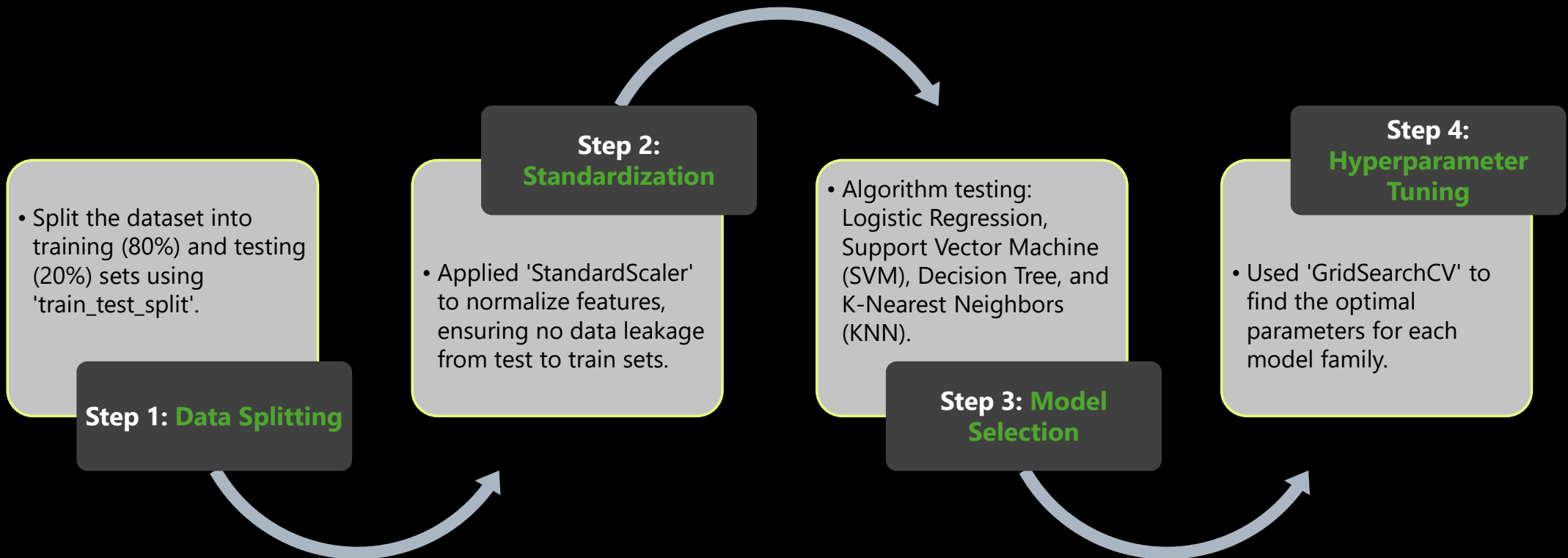
## Interactive Visualization (Folium & Dash)

- Geo-spatial Analysis: Used Folium to build interactive maps and calculate proximity to critical infrastructure.
- Dynamic Dashboards: Developed a Plotly Dash application for real-time success rate filtering.
- Visual Storytelling: Leveraged Seaborn/Matplotlib for multi-variable scatter and bar plots.

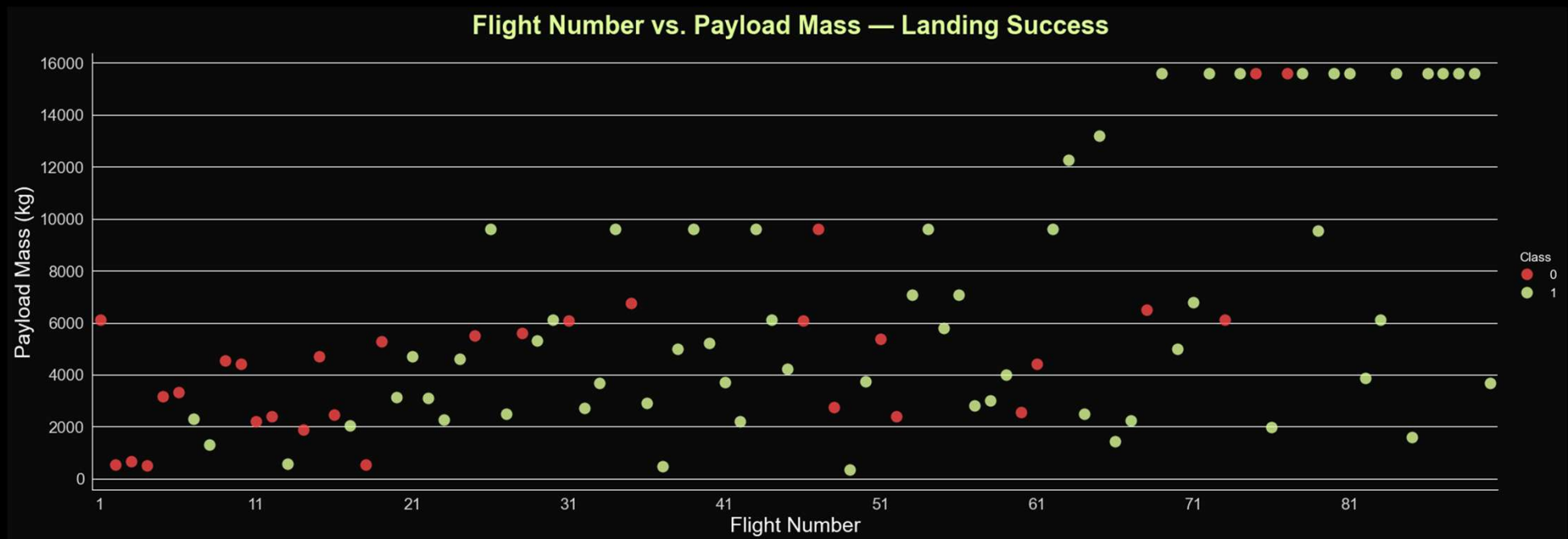


plotly | Dash

# PREDICTIVE ANALYSIS METHODOLOGY



# EDA RESULTS: PAYLOAD VS. SUCCESS



Insight: Reliability has improved over time. Early flights show high failure density, while later flights (post-vol 60) demonstrate near-total landing success, regardless of payload mass.

# EDA RESULTS: SQL STRATEGIC INSIGHTS

Customer Focus 

Total NASA  
Payload

**45,340 KG**

The first successful  
landing on a ground  
pad marked the birth  
of reusability.

Success Milestone 

First Ground  
Success

**2015-12-22**

The first successful  
landing on a ground  
pad marked the birth  
of reusability.

Performance 

Ground Pad  
Rate

**100%**

Zero failures  
observed during  
Landing Pad (RTL)  
maneuvers to date.

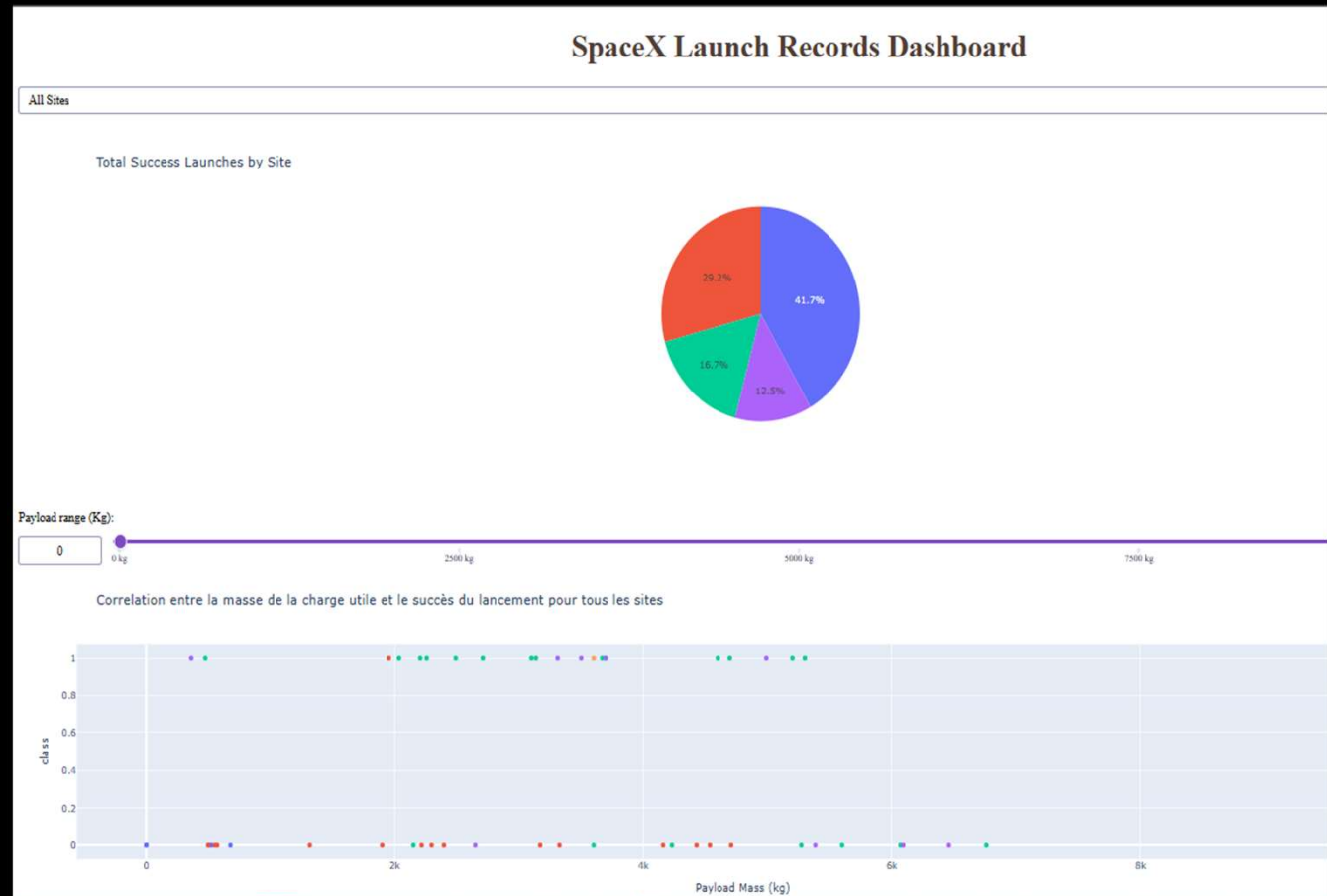
# GEOSPATIAL ANALYSIS: LAUNCH SITE LOGISTICS

- Strategy: Launch pads are intentionally positioned near coastlines (safety) and railways/highways (heavy transport logistics).
- Clusters: High-density launch activity is concentrated at KSC LC-39A and CCAFS SLC-40.

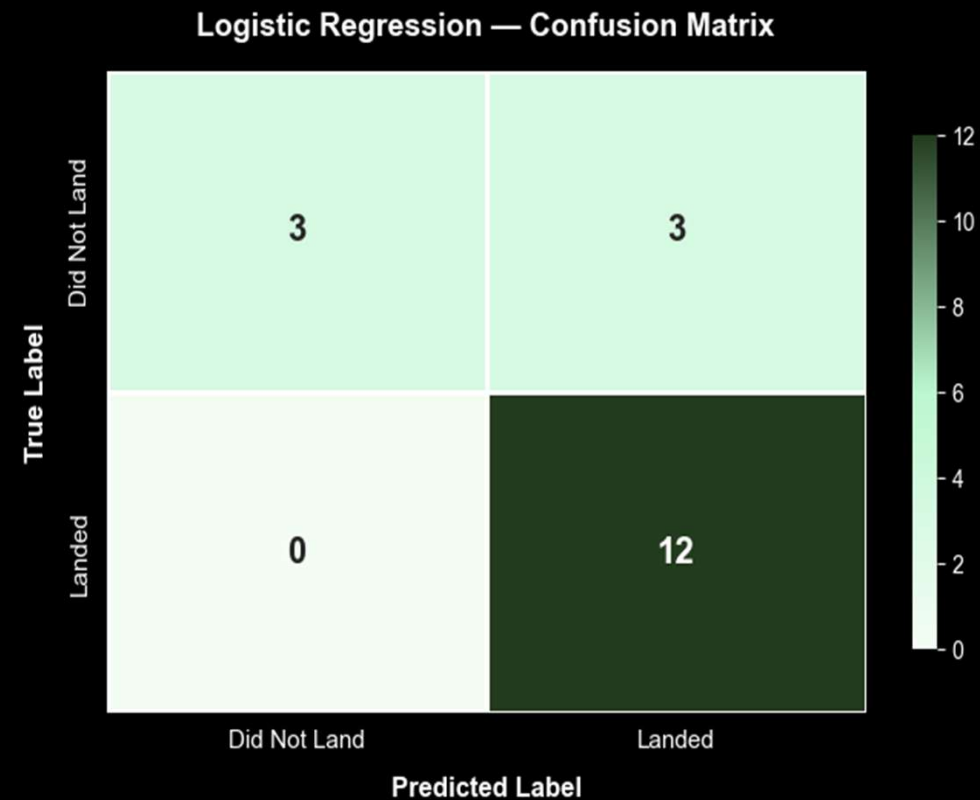


# INTERACTIVE DASHBOARD: PLOTLY DASH

- User Control: Stakeholders can filter success rates by specific launch sites and payload ranges.
- Payload Correlation: The dashboard reveals a "sweet spot" for success between 2,000kg and 6,000kg payload mass.



# PREDICTIVE MODELING: FINAL RESULTS



Selected Model: Logistic Regression (Best balance of accuracy and interpretability).

# CONCLUSIONS & STRATEGIC RECOMMENDATIONS

Technological Lead:  
SpaceX's landing  
success has  
transitioned from  
experimental to  
predictable, with  
83%+ confidence.

Competitive  
Analysis: "Space Y"  
must target the  
4,000-6,000kg  
payload niche where  
booster recovery is  
most optimized.

Economic Impact:  
Reusability reduces  
launch costs by  
~50%, a factor now  
mathematically  
validated through  
our data pipeline.

# FUTURE INNOVATION & MODEL EXPANSION



## Atmospheric Integration

Incorporate real-time weather, wind shear, and launch-day tropospheric data to enhance prediction accuracy.



## Booster Degradation Modeling

Track the aging effect of specific booster core serial numbers over multiple reuses.



## Financial Engine Deployment

Link the model's landing probability directly to programmatic launch insurance pricing tables.